Exercise 1. Find an example of a continuous bounded function $f : \mathbb{R} \to \mathbb{R}$ that does not achieve an absolute minimum or an absolute maximum on $\mathbb{R}$.

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ and $f(x) = (1 + e^{-x})^{-1}$. We need to prove that f is continuous and bounded. If $x_0 \in \mathbb{R}$, then

$$\lim_{x \to x_0} e^x = e^{x_0} \implies \lim_{x \to x_0} (1 + e^{-x}) = 1 + e^{-x_0} \implies \lim_{x \to x_0} f(x) = f(x_0)$$

and

$$0 < e^{-x_0} < \infty \implies 1 < 1 + e^{-x_0} < \infty \implies 0 < f(x_0) < 1.$$

We want to prove that f is strictly increasing. If $x < y$, then

$$e^x < e^y \implies e^{-x} > e^{-y} \implies 1 + e^{-x} > 1 + e^{-y} \implies f(x) < f(y).$$

We need to prove that f achieves neither an absolute minimum nor an absolute maximum. If $y_0 \in \mathbb{R}$, then

$$0 < f(y_0 - 1) < f(y_0) < f(y_0 + 1) < 1.$$

□

Exercise 2. Show that $f : (c, \infty) \to \mathbb{R}$ for some $c > 0$ and defined by $f(x) := 1/x$ is Lipschitz continuous.

Proof. Let $S > 0$ and $f : (S, \infty) \to \mathbb{R}$. Choose $K = S^{-2} > 0$. If $x, y \in (S, \infty)$ and $f(x) = x^{-1}$, then

$$S < x, y < \infty \implies S^2 < xy < \infty \implies 0 < \frac{1}{xy} < K$$

and

$$|f(x) - f(y)| = \left| \frac{1}{x} - \frac{1}{y} \right| = \frac{|x - y|}{xy} < K|x - y|.$$

□

Exercise 3. Show that $f : (0, \infty) \to \mathbb{R}$ defined by $f(x) := \sin(1/x)$ is not uniformly continuous.

Proof. Let $f : (0, \infty) \to \mathbb{R}$ and $f(x) = \sin(x^{-1})$. By the Sequential Characterization of Uniform Continuity, f is not uniformly continuous if and only if $\exists$ sequences $\{x_n\}$ and $\{y_n\}$ in $(0, \infty)$ such that

$$\lim_{n \to \infty} |x_n - y_n| = 0 \ \wedge \ \lim_{n \to \infty} |f(x_n) - f(y_n)| > 0.$$

If

$$x_n = \frac{2}{(1 + 4n)\pi} \in (0, \infty) \implies f(x_n) = \sin\left(\frac{\pi}{2} + 2n\pi\right) \in [-1, 1]$$

and

$$y_n = \frac{2}{(3 + 4n)\pi} \in (0, \infty) \implies f(y_n) = \sin\left(\frac{3\pi}{2} + 2n\pi\right) \in [-1, 1],$$

then

$$\lim_{n\to\infty} |x_n - y_n| = \lim_{n\to\infty} (x_n - y_n) = \lim_{n\to\infty} \frac{4}{(3+16n+16n^2)\pi} = 0$$

and

$$\lim_{n\to\infty} |f(x_n) - f(y_n)| = \lim_{n\to\infty} \left(\sin\frac{\pi}{2} - \sin\frac{3\pi}{2} \right) = 2.$$

□

Exercise 4. Let $S \subset \mathbb{R}$. We say that $f : S \to \mathbb{R}$ is Lipschitz continuous on S if there exists $K \geq 0$ such that for all $x, y \in S$,

$$|f(x) - f(y)| \leq K|x - y|.$$

Prove that if $f : S \to \mathbb{R}$ is Lipschitz continuous on S then f is uniformly continuous on S.

Proof. Let $S \subset \mathbb{R}$ and $f : S \to \mathbb{R}$. Assume, for the purpose of proof, $\exists K \geq 0$ such that

$$\forall x, y \in S : |f(x) - f(y)| \leq K|x - y|.$$

Fix $\epsilon > 0$. Suppose, for the purpose of casework, $K = 0$. Choose $\delta > 0$. If $x, y \in S$ and $|x - y| < \delta$, then

$$|f(x) - f(y)| \leq K|x - y| < K\delta = 0 < \epsilon.$$

Suppose, for the purpose of casework, $K > 0$. Choose $\delta = K^{-1}\epsilon$. If $x, y \in S$ and $|x - y| < \delta$, then

$$|f(x) - f(y)| \leq K|x - y| < K\delta = \epsilon.$$

□

Exercise 5.

(a) Prove that $f(x) = \sin x$ is Lipschitz continuous on $\mathbb{R}$.

(b) Prove that $f(x) = x^{1/3}$ is uniformly continuous on $[0, 1]$ and is not Lipschitz continuous on $[0, 1]$.

Proof. Let $S \subset \mathbb{R}$ and $f : S \to \mathbb{R}$.

(a) Choose $S = \mathbb{R}$ and $f(x) = \sin x$. By the Sine Difference-Product Identity,

$$\begin{aligned} \forall x, y \in S : |f(x) - f(y)| &= |\sin x - \sin y| \\ &= \left| 2\sin\left(\frac{x-y}{2}\right)\cos\left(\frac{x+y}{2}\right) \right| \\ &= 2\left|\sin\left(\frac{x-y}{2}\right)\right|\left|\cos\left(\frac{x+y}{2}\right)\right| \\ &\leq 2\left|\sin\left(\frac{x-y}{2}\right)\right| \leq 2\left|\frac{x-y}{2}\right| = |x-y|. \end{aligned}$$

(b) Choose $S = [0, 1]$ and $f(x) = x^{1/3}$. By the Algebraic Difference of Cubes Identity,

$$\begin{aligned}\forall x, y \in S \setminus \{0\} : |f(x) - f(y)| &= |x^{1/3} - y^{1/3}| \\ &= |x^{1/3} - y^{1/3}| \cdot \frac{|x^{2/3} + x^{1/3}y^{1/3} + y^{2/3}|}{|x^{2/3} + x^{1/3}y^{1/3} + y^{2/3}|} \\ &= \frac{|x - y|}{x^{2/3} + x^{1/3}y^{1/3} + y^{2/3}} \\ &< \frac{|x - y|}{x^{2/3} + y^{2/3}} < \frac{|x - y|}{|x - y|^{2/3}} = |x - y|^{1/3}.\end{aligned}$$

Fix $\epsilon > 0$ and $\delta = \epsilon^3$. If $x, y \in S \setminus \{0\}$ and $|x - y| < \delta$, then

$$|f(x) - f(y)| < |x - y|^{1/3} < \delta^{1/3} = \epsilon.$$

Fix $K \geq 0$. If $K = 0$, then $\exists x, y \in S$ such that

$$x, y > 0 \implies x^{2/3} + x^{1/3}y^{1/3} + y^{2/3} > K;$$

hence,

$$|f(x) - f(y)| = \frac{|x - y|}{x^{2/3} + x^{1/3}y^{1/3} + y^{2/3}} > K|x - y|.$$

If $K > 0$, then $\exists x, y \in S$ such that

$$x, y < \sqrt{\frac{K^3}{27}} \implies x^{1/3}, y^{1/3} < \sqrt{\frac{K}{3}} \implies x^{2/3} + x^{1/3}y^{1/3} + y^{2/3} < K;$$

hence,

$$|f(x) - f(y)| = \frac{|x - y|}{x^{2/3} + x^{1/3}y^{1/3} + y^{2/3}} > K|x - y|.$$

□

Exercise 6. Let $R \in \mathbb{R}$, and let $f : [R, \infty) \to \mathbb{R}$. We say that $f(x)$ converges to λ as $x \to \infty$ if for every $\epsilon > 0$ there exists $M \geq R$ such that for all $x \geq M$ we have $|f(x) - \lambda| < \epsilon$. We write $f(x) \to L$ as $x \to \infty$ or

$$\lim_{x \to \infty} f(x) = \lambda.$$

[A similar definition can be formulated for limits as $x \to -\infty$ but we will not do so here.]

(a) Prove that

$$\lim_{x \to \infty} \frac{x^2}{x^2 + 1} = 1.$$

(b) Prove that

$$\lim_{x \to \infty} \sin x$$

does not exist.

Proof. Let $S \in \mathbb{R}$ and $f : [S, \infty) \to \mathbb{R}$.

(a) Define $f(x) = x^2(x^2+1)^{-1}$ and $\lambda = 1$. Fix $\epsilon > 0$ and $N = \max\{S, \epsilon^{-1/2}\}$. If $x \geq N$, then

$$|f(x) - \lambda| = \left|\frac{x^2}{x^2+1} - 1\right| = \left|\frac{x^2 - x^2 - 1}{x^2+1}\right| = \frac{1}{x^2+1} < \frac{1}{N^2} \leq \frac{1}{\epsilon^{-1}} = \epsilon.$$

(b) Define $f(x) = \sin x$ and $\lambda \in \mathbb{R}$. Fix $\epsilon_0 = 1$ and $M \geq S$. By the Archimedean Property, $\exists N \in \mathbb{N}$ such that $N > M$. Suppose, for the purpose of casework, $\lambda \geq 0$. If

$$x = -\frac{\pi}{2} + 2N\pi > M \implies f(x) = \sin\left(-\frac{\pi}{2} + 2N\pi\right) = -1,$$

then

$$|f(x) - \lambda| = |-1 - \lambda| = 1 + \lambda \geq \epsilon_0.$$

Suppose, for the purpose of casework, $\lambda < 0$. If

$$x = \frac{\pi}{2} + 2N\pi > M \implies f(x) = \sin\left(\frac{\pi}{2} + 2N\pi\right) = 1,$$

then

$$|f(x) - \lambda| = |1 - \lambda| = 1 - \lambda \geq \epsilon_0.$$

□